Huang, Tzu Cheng

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Education

National Yang Ming Chiao Tung University (NYCU) **Sep 2024 – Aug 2026**
M.S. in Data Science and Engineering, College of Computer Science

National Chung Hsing University (NCHU) **Sep 2020 – Jun 2024**
B.S. in Electrical Engineering and Computer Science (EECS)

Experience

Deep Learning Intern **Jul 2025 – Oct 2025**
Moldex3D

- Refactored an existing **CAE prediction pipeline** to support configurable NN/XGBoost training and inference, removing ~**300 lines of duplicated code**.
- Automated training and inference across **29 CAE simulation cases**, adapting to **features specific to each case** and preventing individual failures from interrupting batch runs.
- Automated **R^2 aggregation, prediction visualization, and PowerPoint reporting**, reducing manual result consolidation and accelerating experiment iteration.

Projects

Specification-Driven LLM Framework for RL Environment Generation **Sep 2025 – Jun 2026**
Master's Thesis

- Developed an LLM framework coordinating multiple agents that transforms natural-language game rules into formal specifications, tests, and executable reinforcement learning environments.
- Designed a **verifiable functional environment architecture** using pure functions, immutable persistent state, deterministic validation, and local repair guided by traceability.
- Evaluated across game and software engineering benchmarks, achieving a **25.3% APR improvement** over the best baseline on the complex card game environment.

Computer-Vision Autonomous Agent **Jul 2026 – Present**

- Built an autonomous agent using computer vision, combining perception, temporal state tracking, decision policies, and state-machine control for long-horizon task execution.
- Designed action control that blocks execution under uncertainty with verification on newly captured frames, ownership arbitration, and bounded recovery from uncertain states.
- Developed configurable perception and control modules with explicit uncertainty handling and automated behavioral testing.

Multi AI-Agent Development Workflow **Jul 2026 – Present**

- Built a software engineering workflow coordinating multiple AI agents across specification, repository scouting, implementation, independent review, integration, and cleanup.
- Designed a Git/GitHub lifecycle that blocks unsafe integration using exact commit SHAs, isolated worktrees, review gates, and validated merge authorization.
- Automated task bootstrap, remote AI review, premerge validation, and cleanup with PowerShell and GitHub Actions.

Robot Navigation & Visuomotor Imitation Learning **Sep 2025 – Jan 2026**
Physical AI Course Project

- Built an object goal navigation pipeline in Habitat using RRT* with obstacle inflation and a closed-loop Pure Pursuit controller with dynamic lookahead; reduced **planned path cost by 11.9–36.4% vs. RRT across all evaluated targets**.
- Built and debugged a visuomotor imitation learning pipeline from UMI demonstrations collected in the real world to Isaac Sim closed-loop deployment, achieving **82.67% cup stacking success**.
- Developed reproducible evaluation tooling with trajectory replay, pose alignment, diagnostics across multiple cameras, and quantitative error analysis to isolate policy errors from simulator and coordinate frame mismatches.

Explainable AI for Credit Score Classification **Sep 2025 – Jan 2026**
Data Science Course Project

- Identified target leakage in a published baseline and built the evaluation pipeline to establish a reliable benchmark across XGBoost, CatBoost, GA2M, and Logistic Regression.
- Developed a **unified XAI evaluation framework** to quantitatively compare LIME, SHAP, and intrinsic explanations using faithfulness and robustness metrics across heterogeneous models; **project ranked 1st** in the NYCU data science course project competition.
- Built a counterfactual decision support prototype that generated feature changes with DiCE and integrated an LLM to translate the original counterfactual output into human-readable recommendations.

FastReplay – C++ Replay-Buffer Performance Study for Reinforcement Learning **Feb 2026 – Jun 2026**
Numerical Software Development Course Project

- Built an atomic SPSC **ring buffer** prototype for RL experience replay, **exposing C++ memory to NumPy through pybind11** and implementing random sampling constrained by the current buffer head to prevent stale reads after buffer wraparound.
- Profiled microbenchmarks and full **Stable-Baselines3** training runs; achieved **12.4× faster** sequential batch reads vs. NumPy copy, but found replay buffer operations accounted for only **~6% of training time**, showing that replay-buffer optimization was not the primary bottleneck.

DocCenter **Feb 2025 – Jun 2025**

Cloud Native Development Course Project

- Implemented a versioned document review workflow in NestJS/TypeORM, using transactions to coordinate draft submission, reviewer assignment, and review-state updates.
- Built GitHub Actions CI to run backend tests and publish coverage reports on every push.

Reinforcement Learning Coursework **Sep 2024 – Jan 2025**

Coursework

- Implemented TD learning with an n-tuple value network for 2048, achieving a **96.4% win rate for reaching the 2048 tile over 1,000 games**.
- Developed a PPO racing agent using visual observations with frame stacking and 32 parallel simulators; **ranked 5th of 84** in the final competition.

Numerical Software Development **Feb 2026 – Jun 2026**

Coursework

- Implemented numerical kernels and Python/C++ extensions using C++, MKL/BLAS, pybind11, and CMake, including tiled matrix multiplication and direct NumPy views over C++ memory; benchmarked MKL matrix multiplication at **19.7× faster than a naive implementation**.

Deep Learning Coursework **Feb 2025 – Jun 2025**

Coursework

- Implemented **U-Net/ResNet34-U-Net, MaskGIT, CVAE, and conditional DDPM** across segmentation, video prediction, and image generation; achieved **98.6% evaluator accuracy** on conditional generation.

Skills

Languages: Python, C/C++, JavaScript, SQL, Shell
AI / ML: PyTorch, Machine Learning, Deep Learning, Reinforcement Learning, Computer Vision
Robotics / Simulation: Habitat, Isaac Sim
Frameworks & Data: Flask, React, MySQL, PostgreSQL, MS SQL
DevOps & Engineering: Git, GitHub Actions, Docker, Linux, Git worktrees, CI/CD

Certifications

Power Platform Fundamentals (PL-900)

Microsoft Certified